NATIONAL INSTITUTE OF TECHNOLOGY PATNA
MID SEMESTER EXAMINATION July-Dec 2021

SUBJECT: INTRODUCTION TO COMPUTING
SUBJECT CODE: CS15101/ CS1301
SECTION: ECE A & B/ CE

DURATION: 2 Hrs
FULL MARKS: 30
SEMESTER: 1st

Attempt all questions.

		Marks	CO	BL
Q1.	(a) Explain components of computer system in brief. Explain different types of memory in detail.	[4]	CO1	understand
	(b) What is the purpose of header file? Is the use of header file in the program is necessary? Mention some of the library functions with corresponding header file.	[2]	CO1	understand
Q2.	What are the three iterative constructs used in C? Explain with syntax. Write a program to print factorial of a number using all three constructs	[6]	CO2	Inference
Q3.	Find error (if any) and after correction give output with explanation. Assume code includes all pre-processor directives and main function. **a)** `char c='y'; switch(C) { case 'Y': printf("Yes/No"); case 'N': printf("No/Yes"); break; default: printf("Program is in default case"); }` **b)** `{ int i=3; for(I- -; i<7; i=7) Printf("%d", i++); return 0; }` **c)** `{ int i=5; int *p; P=&i; printf("%d%d%u",*p,*(&i),*(*(&p))); }`	[3*2= 6]	CO2	Inference
Q4.	Distinguish between the following with example. a) Break and Continue statement b) Prefix operators and Postfix operators c) Implicit and Explicit type conversion	[3*2= 6]	CO2	Inference
Q5.	Write a C language program to read n numbers in an array and split the array into two arrays even and odd, such that the array even contains all the even numbers and other is odd. So the output will be— (e.g. Original array is 7,9,4,6,5,3,2,10,18 Odd array is 7,9,5,3 Even array is 6,2,10,18)	[6]	CO3	Inference

Code for Q3:

a)
```
char c='y';
switch(C)
{   case 'Y': printf("Yes/No");
    case 'N': printf("No/Yes"); break;
    default: printf("Program is in default case");
}
```
b)
```
{ int i=3;
  for(I- -; i<7; i=7)
  Printf("%d", i++);
  return 0;
}
```
c)
```
{
  int i=5;
  int *p;
  P=&i;
  printf("%d%d%u",*p,*(&i),*(*(&p)));
}
```

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

ASHOK RAJPATH, PATNA—800 005, (BIHAR)

DEPARTMENT OF ELECTRONCS AND COMMUNIATION ENGINEERING

MID-TERM EXAMINATION

Subject: Elements of Electronics Engineering (EC15102)

Course: B.Tech (ECE), Batch: ECE-1 Session: 2021-22 (Jan. –March 2022)
Date of Exam: 09-03-2022, Time: 02:00 P.M. to 04:00 P.M. (2 HRS), MM: 30

Q. 1	Consider the network given in Fig. 1. Sketch the output voltage V_0 for the given input V_i, The given diode is Si diode having cut-in voltage 0.7 V.	5	**CO-1**

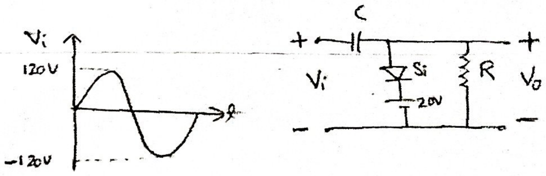

Fig. 1

Q. 2	Consider the network given in Fig. 2. Sketch the output voltage V_0 for the given input V_i, The given diodes are ideal and the each resistance is of 2.2 kΩ.	5	**CO-1**

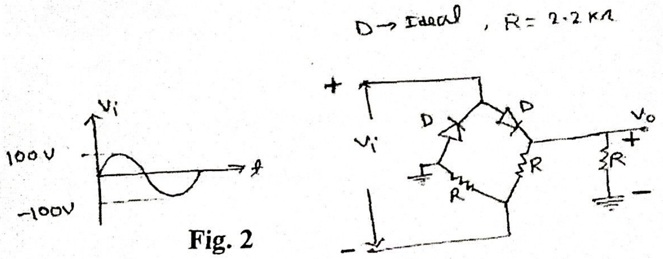

Fig. 2

Q.3	Describe the built-in potential in a p-n junction diode along with its derivation with the help of energy band diagram.	5	**CO-1**
Q. 4	Consider the network given in Fig. 3. Determine I_B, I_E, V_B, and V_C	5	**CO-2**

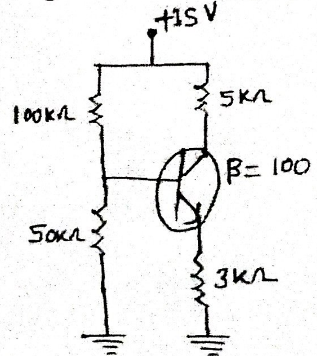

Fig. 3

Q. 5	Represent the comparative analysis of the I-V characteristics (input as well as output) of the npn BJT in all of the three configurations, *i.e.*, CBC, CEC, and CCC.	5	**CO-2**
~~Q. 6~~	~~Describe the Early effect in CBC and CEC by considering npn BJT. Deduce the relation~~ among α, β, and γ. Describe the significance of I_{CBO} and I_{CEO}.	~~5~~	~~**CO-2**~~

NATIONAL INSTITUTE OF TECHNOLGY PATNA

Department of Mathematics

Mid Semester Examination, March -2022

Subject: Mathematics-I (MA15102)

Course: B.Tech ECE -I
Sem :- 1st

Full Marks: 30
Time :2 hours

Answer any four questions one of equal value

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1. (a) Prove that $y_n = m^n a^{mx} (\log a)^n$ for $y = \log(ax+b)$

   (b) If $y^{\frac{1}{m}} + y^{\frac{-1}{m}} = 2x$ then prove that $(1-x^2)y_n = \Sigma(zn+1)xy_{n+1} - (x^2-m^2)y_n = 0.$

   (c) If $u = f(x)\ where\ \ r^2 = x^2 + y^2$, then p.t $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r).$

2. (a) p.t $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}$,Where $x = ucos\alpha - vsin\alpha; y = usin\alpha + vcos\alpha$

   (b) Find the dimensions of a rectangular box of maximum capacity where surface area is given when

   (i) box is open at the top (ii) box is closed.

3. (a) Solve :- (i) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x\ cosx$

   (ii) By variation of parameters solve $\frac{d^2y}{dx^2} + 4y = tan2x$

4. Using elementary transformation find

   (i) Inverse of $\begin{bmatrix} 2 & 1 & -1 & 2 \\ 1 & 3 & 2 & -3 \\ -1 & 2 & 1 & -1 \\ 2 & -3 & -1 & 4 \end{bmatrix}$ by Gauss- Jordan Method

   (ii) Reduce to normal (1st canonical) form and find the rank of $\begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix}$

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